

EvalMaxSAT 2024

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EvalMaxSAT¹ is a MaxSAT solver written in modern C++ language mainly using the Standard Template Library (STL). The solver is built on top of the SAT solver CaDiCaL [1], but any other SAT solver can easily be used instead. EvalMaxSAT is based on the OLL algorithm [2] originally implemented in the MSCG MaxSAT solver [3], [4] and then reused in the RC2 solver [5]. For a general description of how EvalMaxSAT functions, please refer to [6].

The 2024 version of EvalMaxSAT includes a few minor changes:

- To obtain an initial upper bound, only the NuWLS solver [7] is used for 3 minutes.
- Searching for a solution with SCIP is limited to 400 seconds and only if the maximum weight for every soft clause is under 500 million.
- The formula is preprocessed with an adaptation of the SBVA tool [8] if it contains fewer than 100 million hard clauses, and this preprocessing is limited to a maximum of 3 minutes.
- During the search for an unsat core, found but discarded cores are saved to be reconsidered in subsequent iterations without the need to rediscover them.

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¹<https://github.com/FlorentAvellaneda/EvalMaxSAT>